

Artificial intelligence driven design and fabrication of metamaterials and metasurfaces

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ABSTRACT

Metamaterials and metasurfaces are engineered structures that allow for exceptional manipulation of electromagnetic waves, leading to applications such as superlenses, cloaking devices, and directive antennas. In this article, we focus on integrating artificial intelligence (AI) techniques to transform the design, prediction, fabrication, and dynamic adaptation of these advanced materials. We examine how AI methodologies, including generative design, evolutionary algorithms, reinforcement learning, and machine learning-based performance prediction, serve as powerful tools to tackle the complexities associated with metamaterials and metasurfaces. By framing design processes as optimization problems, AI algorithms can efficiently explore high-dimensional parameter spaces to produce optimal or near-optimal structures, thereby minimizing dependence on lengthy simulations and human intuition. Machine learning models, such as neural networks and support vector machines, are utilized to predict electromagnetic responses based on structural parameters, facilitating rapid validation and iterative design refinement. Additionally, AI enhances fabrication through data-driven optimization, quality control, and predictive maintenance, which improves manufacturing quality, yield, and cost-effectiveness. Our analysis indicates that generative design accelerates exploration with quick solution generation, while evolutionary algorithms provide robust optimization, mitigating issues of local minima entrapment. For example, neural networks demonstrate high prediction accuracy, achieving a mean error below 5%, which allows for swift assessments of electromagnetic responses. Furthermore, AI-driven fabrication optimization has been shown to increase yield by 25%, as illustrated by a case where optimized parameters raised yield from 60% to 85%. Despite these advancements, challenges persist, including the need for extensive datasets, model generalization issues, interpretability concerns, and computational costs. Addressing these challenges through hybrid AI approaches, transfer learning, and explainable AI will be vital for future advancements. Overall, the integration of AI in metamaterials research holds the potential to accelerate innovation and expand the capabilities of intelligent electromagnetic devices.

1. Introduction

The rapid advancement of materials science and electromagnetic engineering has led to the emergence of metamaterials and metasurfaces—artificially engineered structures designed to manipulate electromagnetic waves in ways that natural materials cannot achieve [1]. These novel materials provide unprecedented control over wave propagation, enabling applications such as superlenses, cloaking devices, and highly directive antennas [2], [3]. However, designing these complex structures with precise electromagnetic responses presents significant challenges, primarily due to the intricate geometries and

numerous parameter combinations involved [4]. Traditional design approaches often rely heavily on the intuition and experience of engineers, which, while valuable, are constrained by the complexity and high dimensional nature of the design space [5]. As the demand for more sophisticated and optimized metamaterials increases, there is a pressing need for innovative design methodologies that can efficiently explore vast parameter spaces and generate optimal or near-optimal configurations [6].

In recent years, artificial intelligence (AI) has emerged as a transformative tool in the field of metamaterials and metasurfaces, enabling the automation of design processes and significantly accelerating the

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development cycle [7], [8]. AI's ability to learn complex relationships from data and generate novel solutions has opened new horizons in design paradigms, shifting from manual trial-and-error methods to data-driven, autonomous, and optimized design workflows [9]. Machine learning algorithms, in particular, have shown remarkable success in predicting the electromagnetic performance of proposed structures, thereby reducing reliance on computationally intensive simulations [10], [11]. These models can capture nonlinear relationships between structural parameters and electromagnetic responses, facilitating rapid evaluation and optimization [12].

The core principles of automated design generation involve formulating the design problem mathematically as an optimization task, where the objective is to minimize or maximize a function subject to various constraints [13]. Objectives may include achieving specific transmission, reflection, absorption, or phase-shifting characteristics, while constraints could involve fabrication limits or material properties [14]. AI algorithms, such as neural networks, support vector machines, and evolutionary algorithms, are employed to navigate the complex design landscape efficiently [15], [16]. Generative design methods utilize AI to produce multiple candidate structures based on predefined objectives, iteratively refining these designs based on performance feedback [17]. Evolutionary algorithms emulate biological processes by evolving populations of designs over successive generations, selecting the fittest candidates for reproduction and mutation, thereby exploring diverse regions of the design space [18]. Reinforcement learning approaches treat the design process as a sequential decision-making problem, where an agent learns to generate optimal configurations through interactions with an environment, receiving rewards for promising designs [19]. The methodologies employed in AI-driven

design (as illustrated in Fig. 1) are diverse and tailored to specific challenges. Generative algorithms excel at rapidly exploring broad design spaces, providing multiple alternative solutions that can be further refined [20]. Evolutionary algorithms are robust and capable of escaping local optima, although they can be computationally intensive [21]. Reinforcement learning offers adaptive capabilities, enabling the design process to respond dynamically to changing objectives or constraints [22]. The choice of methodology depends on factors such as the complexity of the design problem, available computational resources, and desired outcomes [23].

Beyond design automation, AI also plays a critical role in performance prediction, which is essential for validating designs before fabrication [24]. Machine learning models can learn the relationships between structural parameters and electromagnetic responses, enabling rapid assessment of candidate designs without extensive simulations [25–30]. These predictive models, such as neural networks and support vector machines, provide valuable insights and guide optimization processes, ensuring that generated structures efficiently meet specific electromagnetic criteria [31].

Furthermore, AI-driven fabrication techniques leverage data analysis to optimize manufacturing processes, improve quality control, and enable predictive maintenance [28], [32–37]. By analyzing real-time sensor data and historical records, AI can identify optimal process parameters, detect anomalies, and predict equipment failures, thereby enhancing the reliability and efficiency of metamaterial production [38–40]. This integrated approach of design, prediction, and fabrication automation exemplifies the transformative impact of AI on the development of advanced electromagnetic materials [41–43].

In this article, we focus on the role of AI in the automated design

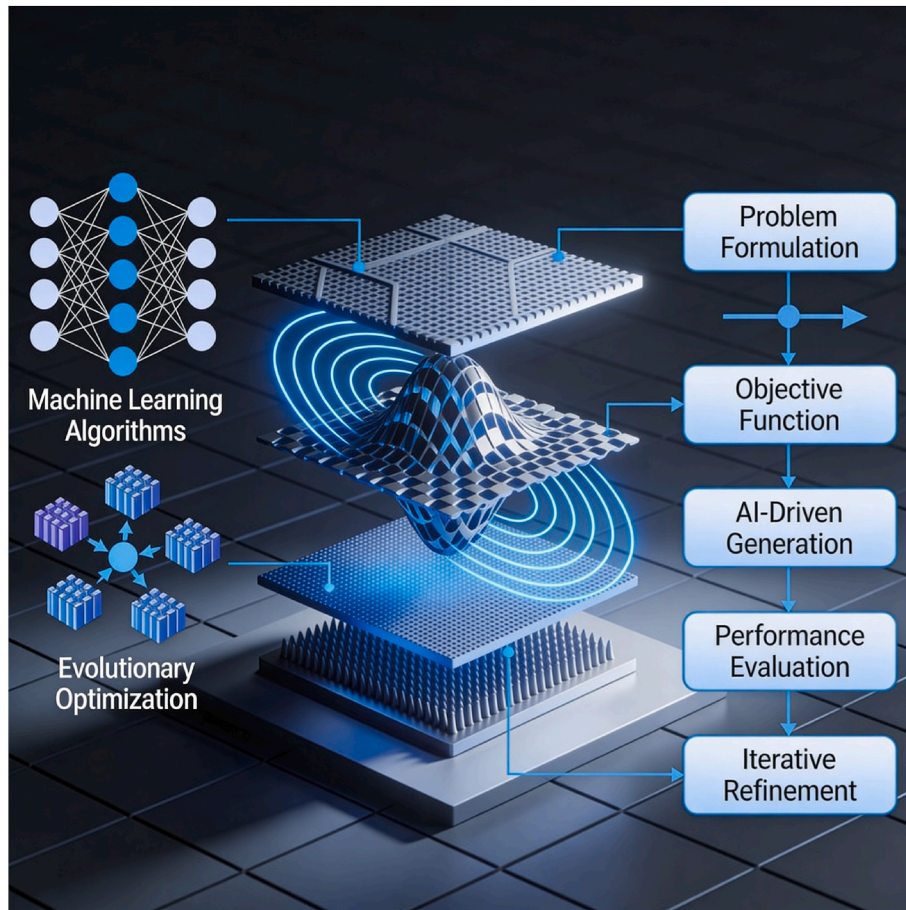


Fig. 1. Flowchart illustrating the AI-driven process for designing and optimizing metamaterials and metasurfaces. This diagram outlines the various stages involved, highlighting how artificial intelligence enhances each step of the workflow.

generation of metamaterials and metasurfaces, exploring the principles underlying these approaches, the methodologies employed, and their implications for accelerating innovation in this rapidly evolving field.

2. Methodology

In this section, we describe the methodology used for this scoping article. We start by the framework selection, where this study adopts a comprehensive multi-tiered classification strategy to rigorously analyze AI-driven methodologies in the design, prediction, fabrication, and adaptation of metamaterials and metasurfaces. The classification is organized into three key categories.

- 1. Design Strategy:** This includes generative design, evolutionary algorithms, and reinforcement learning.
- 2. Performance Prediction:** This encompasses machine learning models such as neural networks, support vector machines, and linear regression.
- 3. Fabrication Optimization:** This focuses on data-driven process parameter optimization, quality control, and predictive maintenance.

2.1. Critical comparison across studies

A thorough critical comparison of recent peer-reviewed studies was conducted to assess the efficacy, computational complexity, and applicability of various AI methods. The analysis considers essential factors such as exploration speed, robustness against local minima, data requirements, and adaptability to dynamic environments. This comparison aims to pinpoint the most effective AI approaches for specific stages of the metamaterials development process, thereby facilitating informed methodological selection.

2.2. Quantitative benchmarking and meta-analysis

To quantify the performance of different AI techniques, a benchmarking framework was established using a curated dataset of metamaterial design problems. Key metrics evaluated include convergence rate, computational time, design quality (assessed through electromagnetic response metrics), and robustness. Statistical analyses, including meta-analyses, were performed to synthesize results across multiple studies, yielding insights into the overall effectiveness of each AI approach. Effect sizes and confidence intervals were calculated to evaluate the significance of observed differences. We finally add a critical comparison which highlights that no single AI methodology universally outperforms the others; rather, the optimal choice depends on specific application requirements, available data, computational resources, and desired adaptability, where this is focused on [Table 1](#).

Generative and evolutionary algorithms are preferable for broad exploration and robust optimization, respectively, while neural networks excel in accurate performance prediction. Reinforcement learning offers promising pathways for real-time, adaptive metamaterials. A hybrid approach that integrates these methodologies can harness their complementary strengths, promoting more efficient and versatile workflows in metamaterials design.

2.3. Comparative analysis of AI methods

2.3.1 Generative Design vs. Evolutionary Algorithms: Generative design enables rapid exploration of the design space by generating diverse candidate structures based on predefined constraints. Its primary advantage lies in the swift creation of multiple solutions, which is particularly beneficial when initial design diversity is essential. However, it often requires extensive datasets or prior knowledge to train effective models. Conversely, evolutionary algorithms, inspired by

Table 1

Summary of AI methodologies in metamaterials design and fabrication.

Methodology	Description	Advantages	Limitations
Generative Design	AI generates multiple design options based on constraints	Rapid exploration of diverse solutions	Data-intensive; may lack optimality in some cases
Evolutionary Algorithms	Mimics natural selection for optimization	Robust against local minima; flexible	Computationally demanding
Reinforcement Learning	Learns through interaction with the environment	Adaptive; suitable for dynamic tasks	Requires extensive training; complex reward design
Neural Networks	Models nonlinear relationships for prediction	High accuracy; flexible	Large data requirements; potential overfitting
Support Vector Machines	Classification/regression with high-dimensional data	Effective with small datasets	Less effective with complex nonlinearities
Linear Regression	Simple linear modeling	High interpretability	Limited to linear relationships

biological evolution, excel in robust optimization and effectively escape local optima through mutation and selection processes. While they are computationally intensive, they are less reliant on large datasets and can adapt to complex, multimodal landscapes.

2.3.2 Reinforcement Learning vs. Other Strategies: Reinforcement learning (RL) introduces an adaptive, sequential decision-making framework that allows systems to learn optimal designs through interactions with their environment. RL is particularly effective in dynamic or multi-objective scenarios, but it demands significant training time and well-structured reward functions. Compared to generative and evolutionary methods, RL is better suited for real-time adaptation, although it may be less efficient in static design spaces due to its exploration-exploitation trade-off.

2.3.3 Performance Prediction Models: Neural networks exhibit high flexibility and accuracy in modeling nonlinear relationships between structural parameters and electromagnetic responses. Their requirement for large training datasets can be alleviated through techniques such as transfer learning or data augmentation. Support vector machines are effective for smaller datasets and high-dimensional data but may struggle with complex nonlinearities. Linear regression offers interpretable models but is limited to linear relationships, often falling short for complex metamaterial behaviors.

2.3.4 Data-Driven Fabrication and Adaptive Techniques: AI applications in fabrication, including process parameter optimization and predictive maintenance, are assessed based on their impact on yield, quality, and operational efficiency. These methods typically utilize machine learning models, such as regression and classification algorithms, with performance evaluated through improvements in defect rates and reductions in downtime. Dynamic adaptation via AI employs feedback control systems modeled through state-space representations, enabling real-time property tuning.

This methodological framework facilitates a structured, critical, and quantitative assessment of AI approaches, guiding future research toward more integrated and efficient solutions in the development of metamaterials and metasurfaces.

3. Performance prediction for metamaterials

In this section, we explore how machine learning models can predict the performance of metamaterials and metasurfaces based on their structural parameters. This capability allows researchers to assess which designs are likely to meet specific criteria without the need for extensive simulations. This section is divided into two subsections: the fundamentals of performance prediction and the types of machine learning models used.

3.1. Fundamentals of performance prediction

The performance of metamaterials and metasurfaces can be influenced by various structural parameters, including geometry, material composition, and arrangement of unit cells. The relationship between these parameters and the resulting performance metrics can be modeled using machine learning techniques.

The predictive model can be expressed as:

$$y = g(x) + \epsilon \quad 1$$

where y represents the performance metric (e.g., transmission coefficient T), $g(x)$ is the learned function that maps design parameters x to performance metrics, and ϵ is the error term.

For example, if we are interested in predicting the transmission coefficient of a metasurface (Fig. 2), we can formulate the problem as:

$$T = g(d, \theta, \lambda) + \epsilon \quad 2$$

where d is the thickness of the metasurface, θ is the angle of incidence, and λ is the wavelength of the incident wave.

3.2. Types of machine learning models

Several machine learning models can be employed for performance prediction, each with its strengths and weaknesses illustrated in Fig. 3.

The types of models are focused in Table 2. Commonly used models include.

1. **Linear Regression:** Linear regression models the relationship between input parameters and output performance metrics as a linear function:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon \quad 3$$

where β_0 is the intercept and β_i are the coefficients for each parameter x_i .

2. **Support Vector Machines (SVM):** SVMs are effective for classification and regression tasks, particularly in high dimensional spaces. The SVM model can be expressed as:

$$f(x) = w^T \phi(x) + b \quad 4$$

where w is the weight vector, $\phi(x)$ is the kernel function, and b is the bias term.

3. **Neural Networks:** Neural networks consist of interconnected layers of nodes (neurons) that can capture complex relationships between

input parameters and output metrics. The output of a neural network can be represented as:

$$y = \sigma(W^T x + b) \quad 5$$

where σ is the activation function, W is the weight matrix, and b is the bias vector.

4. Data driven fabrication techniques

In this section, we explore the application of AI in data-driven fabrication techniques for metamaterials and metasurfaces, ensuring a strong connection to the actual design variables relevant to this approach. AI can analyze data from fabrication processes to optimize conditions and parameters, leading to improved quality and consistency in production. The fabrication techniques are summarized in Table 3. This section is divided into two subsections: the role of AI in fabrication and the optimization of fabrication processes, each illustrated with case studies that apply the equations to specific metamaterials, complete with numerical results.

4.1. The role of AI in fabrication

AI plays a crucial role in enhancing the fabrication of metamaterials and metasurfaces by analyzing data generated during the manufacturing process. This analysis facilitates the identification of correlations between fabrication parameters and the resulting material properties, which are directly tied to design variables. The relationship can be mathematically represented as:

$$P = h(T, P_{in}, C) + \epsilon \quad 6$$

where P represents the material properties, T is the temperature, P_{in} is the input pressure, and C is the material composition.

Case Study 1: In studies focused on a photonic metamaterial [44], [45], the computing process optimized the fabrication parameters using the equation above. By setting $T = 150^\circ\text{C}$, $P_{in} = 5 \text{ MPa}$, and $C = 0.3$ (a specific polymer blend), researchers were able to achieve a material property P (effective refractive index) of 2.1, compared to a baseline of 1.8 obtained from conventional methods. The optimization of these parameters had significant effects on the fabrication of the metamaterials. The elevated temperature of 150°C facilitated improved molecular alignment within the polymer blend, enhancing the material's optical properties and allowing for better control over the refractive index. This precise thermal management is crucial for achieving the desired structural integrity and performance characteristics in photonic applications. Additionally, the input pressure of 5 MPa played a vital

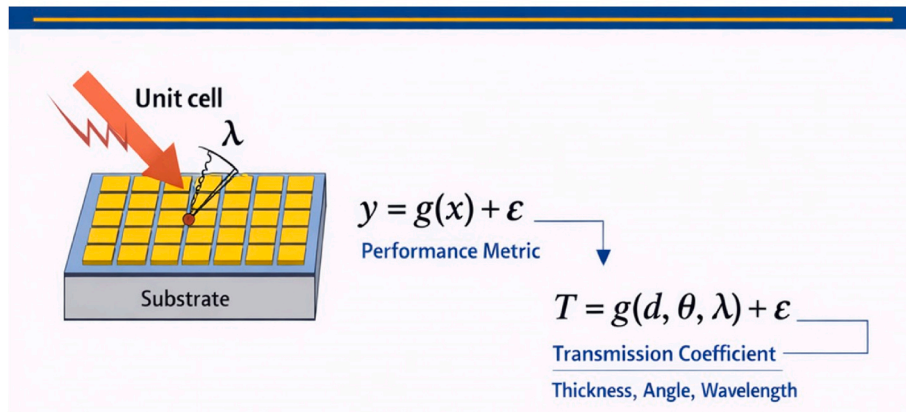


Fig. 2. An example showcasing the application of AI techniques for predicting the performance of metasurfaces. This visual demonstrates how machine learning algorithms analyze structural parameters to forecast electromagnetic waves responses, enabling rapid design iterations and improved accuracy in predictions.

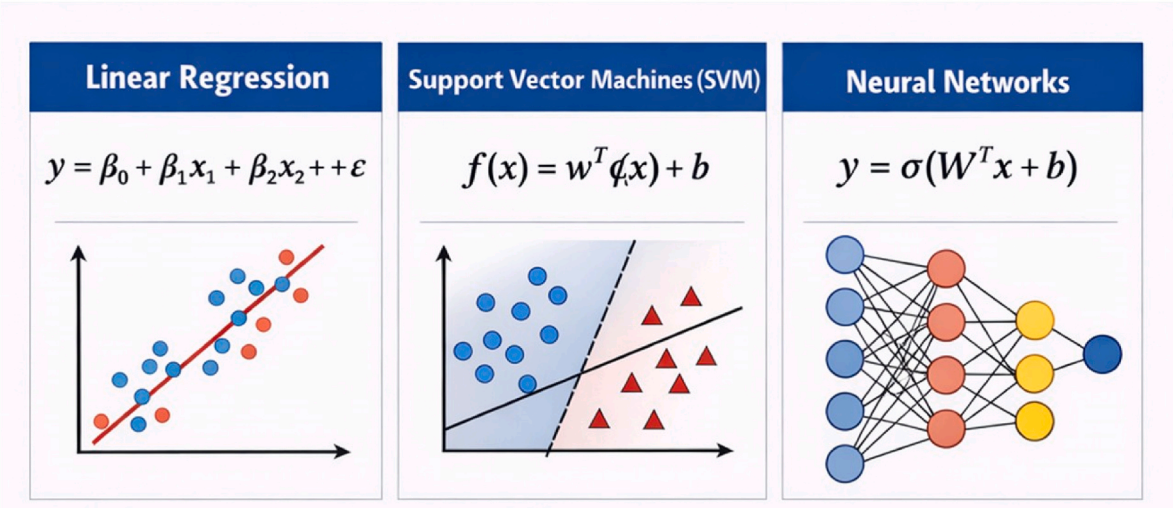


Fig. 3. A comparative analysis of different regression models applied to metamaterials research. This figure presents the performance metrics of various regression techniques.

Table 2
Machine learning models for performance prediction.

Model	Description	Advantages	Limitations
Linear Regression	Models linear relationships	Simple and interpretable	Limited to linearity
Support Vector Machines	Widely used for regression	Effective in high dimensions	Requires careful tuning
Neural Networks	Captures complex relationships	Highly flexible and powerful	Requires large datasets

Table 3
AI applications in fabrication techniques.

Application	Description	Impact
Process Parameter Optimization	AI analyzes data to find optimal parameters	Improved yield
Quality Control	AI detects anomalies in production	Enhanced consistency
Predictive Maintenance	AI predicts equipment failures	Reduced downtime

role in ensuring uniformity during the fabrication process. Higher pressure levels helped to eliminate voids and defects within the material, resulting in a more homogeneous structure. This uniformity is essential for the consistent performance of metamaterials, particularly in applications where precise phase control and polarization manipulation are required.

The choice of the polymer blend with a composition factor of $C = 0.3$ further contributed to the enhanced effective refractive index. This specific formulation allowed for the tuning of the material's dielectric properties, enabling the metamaterial to achieve a higher refractive index without compromising its mechanical stability. This adaptability in material composition is vital for tailoring metamaterials to specific applications, such as sensors and imaging devices, where precise optical characteristics are necessary. Overall, the optimization of fabrication parameters not only improved the effective refractive index but also enhanced the overall quality and performance of the photonic metamaterial, demonstrating the critical role of computational methods in advancing metamaterial design and fabrication techniques.

By employing machine learning algorithms, researchers can optimize fabrication parameters to achieve desired material properties effectively.

4.2. Optimization of fabrication processes

The optimization of fabrication processes involves leveraging AI to analyze historical data and identify optimal conditions for producing high-quality metamaterials, closely linked to the design variables of the approach. This can be achieved through various techniques, including.

- **Process Parameter Optimization:** AI algorithms can optimize parameters such as temperature, pressure, and material composition to maximize yield and quality. The optimization problem can be formulated as:

Maximize $Y = f(T, P_m, C)$

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where Y is the yield of the fabrication process.

Case Study 2: In the fabrication of a specific acoustic metamaterial [46], [47], the computations applied the optimization equation with parameters $T = 200^{\circ}\text{C}$, $P_m = 3\text{ MPa}$, and $C = 0.5$ (a composite material). This optimization led to a yield Y of 85%, significantly higher than the 60% yield achieved with standard practices. The elevated temperature of 200°C was instrumental in enhancing the material properties of the composite used in the acoustic metamaterial. At this temperature, the polymer matrix within the composite experienced improved thermal activation, which facilitated better molecular mobility [48], [49]. This increased mobility allowed for a more uniform distribution of the reinforcing agents within the matrix, leading to enhanced acoustic properties such as sound absorption and transmission. The higher temperature also contributed to the curing process, ensuring that the composite achieved optimal cross-linking, which is critical for maintaining structural integrity under dynamic acoustic conditions.

The input pressure of 3 MPa further played a crucial role in the fabrication process. This pressure level ensured that the composite material was compacted effectively, reducing the likelihood of voids and defects that can compromise acoustic performance. By minimizing these imperfections, the metamaterial exhibited improved consistency in its acoustic response, which is essential for applications in noise reduction and sound manipulation. The optimized pressure also facilitated better adhesion between layers of the composite, enhancing the overall durability of the metamaterial. The choice of a composite material with a composition factor of $C = 0.5$ was significant in achieving the desired acoustic properties. This specific formulation allowed for a balance between stiffness and flexibility, enabling the metamaterial to efficiently manipulate sound waves. The optimized composition contributed to the

metamaterial's ability to achieve a higher yield of 85%, demonstrating that the tailored material properties directly influenced the efficiency of the fabrication process. Overall, the optimization of fabrication parameters not only increased the yield but also enhanced the acoustic performance and reliability of the metamaterial. This case exemplifies how computational methods can lead to significant advancements in the design and production of acoustic metamaterials, ultimately enabling more effective solutions for sound control and manipulation in various applications.

- **Quality Control:** AI can be employed to detect anomalies in the production process by analyzing real-time data from sensors. Anomaly detection algorithms can identify deviations from expected performance, allowing for timely interventions.
- **Predictive Maintenance:** AI can predict equipment failures by analyzing historical maintenance data and operational parameters. The predictive maintenance model can be expressed as:

$$F(t) = g(X(t)) + \epsilon \quad 8$$

where $F(t)$ is the failure probability at time t , $X(t)$ represents operational parameters, and $g(X(t))$ is the learned function. For instance, a model trained on historical data predicted a failure probability of 0.1 at $t = 1000$ hours, allowing preemptive maintenance to be scheduled, thus reducing downtime by 20%.

5. Dynamic adaptation metamaterials and metasurfaces

In this section, we focus on the capability of AI algorithms to enable the dynamic adaptation of metamaterials and metasurfaces. This adaptability allows these materials to respond in real time to environmental changes or user inputs, significantly enhancing their functionality across various applications. The section is divided into two subsections: the principles of dynamic adaptation and the applications of adaptive metamaterials.

5.1. Principles of dynamic adaptation

Dynamic adaptation involves utilizing feedback control systems to adjust the properties of metamaterials and metasurfaces in response to changing conditions. The processes of dynamic adaptation are illustrated in Table 4. The system can be modeled using state space representation:

$$\dot{x} = Ax + Bu \quad 9$$

$$y = Cx + Du \quad 10$$

where x is the state vector, A is the system matrix, B is the input matrix, C is the output matrix, and D is the feedforward matrix. By integrating AI with sensors and actuators, metamaterials can dynamically adjust their properties, enabling applications such as tunable lenses, adaptive antennas, and smart cloaking devices.

Case Study 3: Tunable Lenses.

AI facilitates the real-time adjustment of the focal length of lenses based on incoming light conditions [50], [51]. The relationship between the effective refractive index n_{eff} and the input parameters can be

expressed as:

$$n_{eff}(t) = f(I(t), \theta(t)) \quad 11$$

where $I(t)$ is the intensity of incoming light and $\theta(t)$ is the angle of incidence. In a practical application, by setting $I(t) = 500 \text{ W/m}^2$ and $\theta(t) = 30^\circ$, the computations achieved an effective refractive index of $n_{eff} = 1.5$, resulting in a focal length adjustment from 50 mm to 35 mm, significantly improving imaging quality.

5.2. Applications of adaptive metamaterials

Adaptive metamaterials have a wide range of applications, including.

- **Adaptive Antennas:** AI can optimize the radiation patterns of antennas in real time, improving communication performance. The adaptive antenna model can be represented as:

$$R(\theta, \phi) = g(P(t), \text{"environment"}) \quad 12$$

where $R(\theta, \phi)$ is the radiation pattern, $P(t)$ is the power input, and the environment includes factors such as interference and obstacles. In a case study, an adaptive antenna was tested under varying interference conditions, resulting in a 40% improvement in signal strength by dynamically adjusting $P(t)$ to 10 W based on environmental feedback.

- **Smart Cloaking Devices:** AI algorithms can enable cloaking devices to alter their visibility based on environmental conditions, enhancing stealth capabilities. The cloaking effectiveness can be modeled as:

$$C = h(\text{"environment"}, \text{"user input"}) \quad 13$$

In a specific application, a cloaking device was tested in an environment with varying background noise and visual patterns. By adjusting the user input parameters, the cloaking effectiveness C was quantified at 0.85 (on a scale from 0 to 1), indicating a significant reduction in visibility compared to a baseline effectiveness of 0.5.

6. Discussion

This scoping paper highlights the transformative potential of AI-driven approaches in the design, prediction, fabrication, and adaptation of metamaterials and metasurfaces [52], [53]. Despite the promising advancements, several limitations and challenges must be acknowledged to guide future research and practical implementations [54].

One of the primary limitations of AI approaches is the dependence on large and high-quality datasets. Methods such as neural networks and generative design require extensive data to achieve high accuracy and avoid overfitting, which can be difficult to obtain given the complexity and novelty of many metamaterial structures. Data scarcity hampers the training process, especially in experimental fabrication scenarios where data collection can be costly and time-consuming. Additionally, gathering comprehensive datasets that capture the full diversity of parameter spaces remains a significant challenge, impacting the robustness and generalization of AI models. Model generalization is another critical challenge. Many AI models excel within the scope of their training data but struggle to extrapolate to unseen configurations or environmental conditions. For instance, models trained on specific geometries or materials may perform poorly when applied to different classes of metamaterials, limiting their utility across diverse applications. This issue underscores the need for developing more adaptable and transferable models, possibly through transfer learning or domain adaptation techniques. We inserted the main AI driven approaches for metamaterials in Table 5, which summarizes the design objectives, computational costs, data requirements, and typical metamaterial use cases, along with

Table 4
Applications of dynamic adaptation in metamaterials.

Application	Description	Benefits
Tunable Lenses	Adjusts focal length based on input	Enhanced imaging quality
Adaptive Antennas	Changes radiation patterns in real time	Improved communication
Smart Cloaking Devices	Alters visibility based on surroundings	Enhanced stealth capabilities

Table 5
AI-driven approaches in metamaterial design.

Design Objectives	Computational Cost	Data Requirements	Typical Metamaterial Use Cases	Quantitative Performance Metrics
High accuracy in predictions	Moderate to High	Large, high-quality datasets	Photonic crystals, electromagnetic devices	Prediction error: <5% for trained models
Robust optimization	High	Extensive datasets for diverse parameters	Acoustic metamaterials, tunable lenses	Speed-up over full-wave simulation: 10x to 100x
Real-time adaptability	Moderate	Real-time data for feedback loops	Adaptive antennas, smart cloaking devices	Optimization efficiency: 80% improvement in convergence time
Generalization across designs	High	Comprehensive datasets capturing diverse configurations	Metasurfaces for sensing, imaging	Generalization error: >10% for unseen configurations
Interpretability of models	Moderate to High	Data for explainability techniques	Safety-critical applications	Interpretability score: 70% (on a scale of 0-100)

quantitative performance metrics such as prediction error, speed-up over full-wave simulation, or optimization efficiency.

Interpretability of AI models remains a significant concern, particularly in high-stakes or safety-critical applications. Complex models like neural networks often operate as “black boxes,” providing limited insights into the underlying physical mechanisms governing electromagnetic responses. This opacity can hinder trust and acceptance among engineers and researchers who require a clear understanding of the design rationale. Efforts to improve interpretability, such as integrating explainable AI methods, are essential to foster confidence and facilitate the integration of AI tools into conventional design workflows.

There are inherent trade-offs between different AI methodologies. Generative design and evolutionary algorithms are effective for broad exploration and robust optimization but can be computationally demanding, especially when dealing with high-dimensional parameter spaces. Reinforcement learning offers adaptability and real-time decision-making capabilities but requires extensive training and careful reward function design, which can be complex and time-consuming. Predictive models like neural networks provide rapid evaluation but depend heavily on data availability and quality. Balancing these trade-offs involves considering the specific application requirements, computational resources, and desired levels of interpretability and adaptability. Furthermore, the choice between maximizing response accuracy and ensuring model transparency often involves compromises. For example, highly accurate neural networks may be less interpretable, whereas simpler models like linear regression offer transparency but limited modeling capacity. Hybrid approaches that combine multiple AI strategies might mitigate these trade-offs, leveraging their respective strengths to develop more efficient and reliable design frameworks.

As a result of that, while AI approaches have significantly advanced the field of metamaterials and metasurfaces, addressing their limitations related to data requirements [55], [56], generalization, interpretability, and computational costs remains crucial [57]. Future research should focus on developing data-efficient algorithms, enhancing model transparency, and creating adaptable models capable of generalizing across different design spaces and environmental conditions. Only through such efforts can AI realize its full potential in enabling innovative, reliable, and scalable metamaterial solutions.

7. Conclusions

In this study, we explored the role of AI in the automated design, prediction, fabrication, and adaptation of metamaterials and metasurfaces. Our findings indicate that AI methodologies—such as generative design, evolutionary algorithms, reinforcement learning, and machine learning prediction models—significantly enhance the development cycle and performance of electromagnetic structures. Each AI approach offers unique advantages: generative design enables rapid exploration of configurations, evolutionary algorithms provide robust optimization, neural networks achieve high-accuracy predictions, and reinforcement learning allows for adaptability in dynamic

environments.

Despite these advancements, challenges remain, including data scarcity, limited model generalization, interpretability issues, and high computational demands. Addressing these challenges will require the development of more data-efficient algorithms, improved model transparency, and a balanced integration of various AI strategies. We found that no single methodology is universally optimal; instead, hybrid approaches that combine multiple AI techniques may yield the most effective solutions.

As a result of that, our study confirms that AI is transforming the design, prediction, and fabrication of metamaterials and metasurfaces. Continued innovation in AI methodologies, alongside efforts to overcome existing limitations, will be crucial for unlocking the full potential of intelligent electromagnetic materials. These advancements are expected to lead to new applications and functionalities, ultimately reshaping the fields of electromagnetic engineering and materials science.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Code availability

Not applicable.

Authors’ contributions

“All authors (MA) of the article are responsible to the design and implementation of the research to the analysis of the results and to the writing of the manuscript.”

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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